SIMATS ENGINEERING

ASSESSMENT TOOL 1: Agent Design Assessment

COURSE NAME: **ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS** **COURSE CODE:** **MLA01**

COURSE OUTCOME COVERED

CO1: *Apply search algorithms and heuristic techniques to solve state-space search and optimization problems.* (BTL 3)

Assessment Tool Weightage: 55 %

Total Marks: 50 **Time: 60 Mins**
No. of Questions: 1

Annexure A

Agent Design Assessment

Code of Conduct:

I B.Jahnavi / 192325043give me from introduction onwa (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Question	Problem Understanding & AI Concept Identification	Dialogflow Agent Design & Implementation	Problem Formulation & Conversation Flow Design	Application of AI Techniques & Reasoning	Testing, Evaluation & Documentation	Total
Weightage	20%	30%	20%	20%	10%	100%
Q1						

Signature of the Faculty Total Marks Awarded

Scenario

A university wants to develop an **AI-based Student Support Assistant** to provide instant responses to student queries related to course registration, timetable, examination schedules, attendance details, and campus services.

As an AI developer, design and implement a conversational agent using **Dialogflow** that can understand user requests, process intents, and provide appropriate responses.

Q. No.	Criteria to be evaluated	BL	Marks
1	**Problem Analysis and Agent Design:** Analyze the student support problem and explain how Artificial Intelligence can solve the problem. Design the conversational agent by defining its objective, users, environment, inputs, outputs, and expected performance measures.	BL3	10
2	**Dialogflow Agent Development:** Create a Dialogflow agent by designing appropriate Intents, Entities, Training Phrases, Responses, and Contexts. Explain how the agent perceives user queries and generates responses. Provide screenshots of the implementation.	BL5	10
3	**Problem Formulation and Decision Flow:** Represent the chatbot interaction as a problem-solving process by defining the initial state, user goal, possible actions, state transitions, and goal completion conditions. Draw the conversation flow diagram.	BL6	10
4	**Search Strategy Application:** Analyze how search techniques used in AI problem solving can support conversational agents. Apply a suitable search approach (such as Breadth First Search or heuristic-based search) to model intent selection or dialogue navigation and justify your choice.	BL5	10
5	**Testing and Evaluation:** Test the Dialogflow agent with different user queries. Evaluate accuracy, response quality, handling of unknown queries, and suggest improvements.	BL6	10

AI-Based Student Support Assistant using Dialogflow

PROJECT OVERVIEW:

The AI-Based Student Course Registration Assistant is a chatbot developed using Google Dialogflow to help students with course registration and other academic-related queries. The chatbot uses Artificial Intelligence and Natural Language Processing (NLP) to understand what users ask and provide accurate responses. Students can easily get information about registration, credit limits, elective courses, registration status, and course withdrawal without waiting for manual assistance. This system improves communication between students and the institution while providing support at any time.

NEED FOR THE PROJECT :

Many students face difficulties while searching for information about course registration and academic procedures. During registration periods, faculty members and administrative staff receive a large number of repeated questions, making the process slow and time-consuming. A chatbot can solve this problem by providing instant responses, reducing waiting time, and making information easily available.

PROJECT GOALS:

The main goals of this project are:

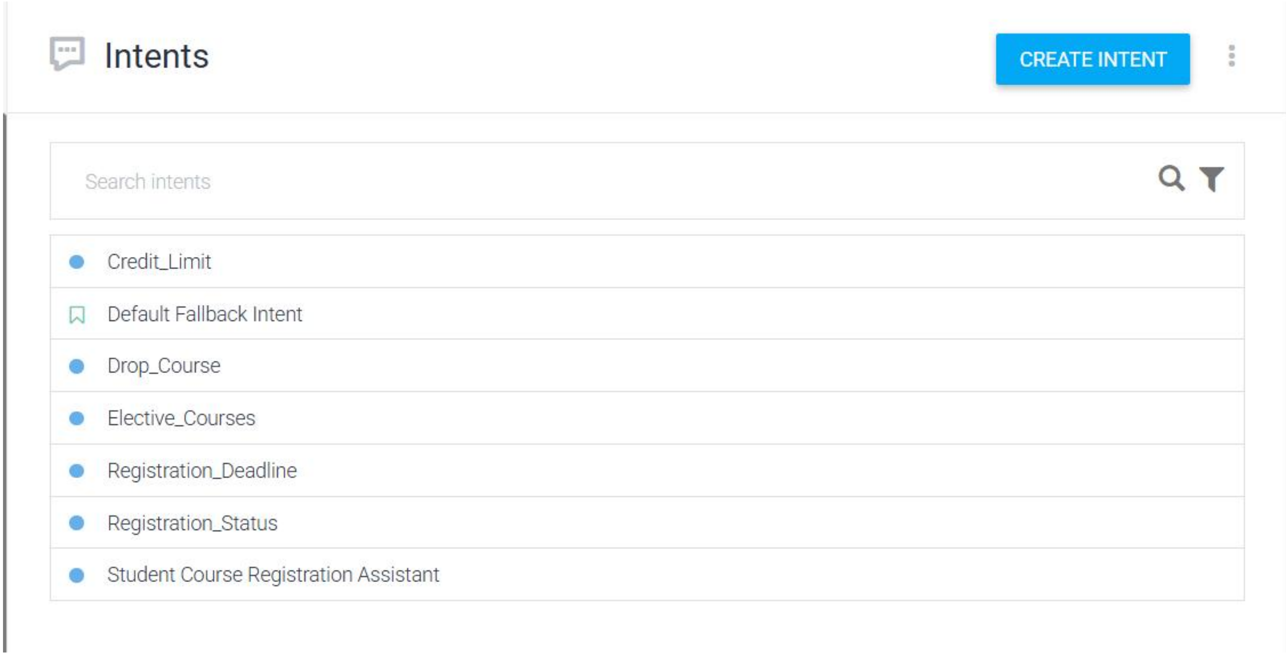
- ✓ To provide quick and accurate answers to students.
- ✓ To reduce the workload of faculty and administration.
- ✓ To improve communication through an automated chatbot.
- ✓ To make academic support available 24×7.
- ✓ To demonstrate the practical use of Artificial Intelligence in education.

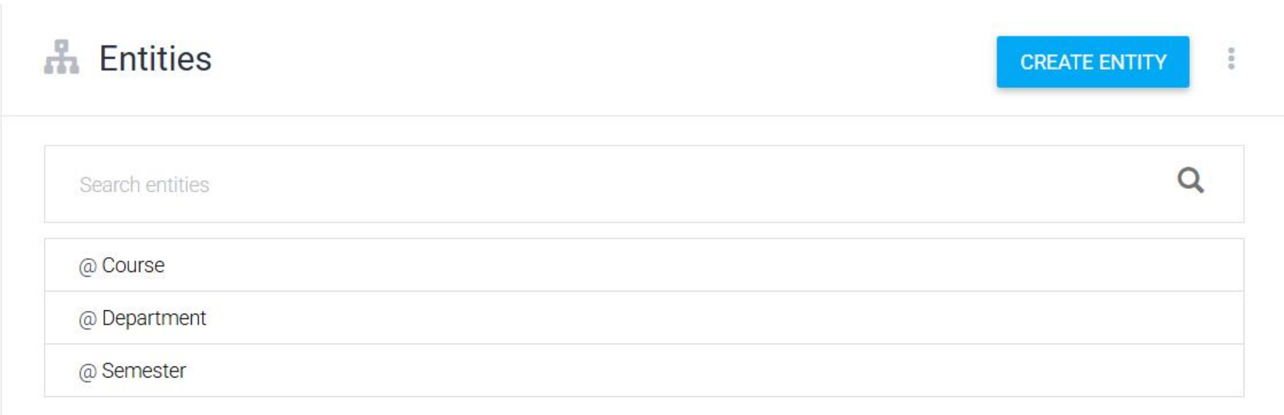
REVIEW OF THE CURRENT STATE :

Currently, students depend on faculty members or administrative offices for information regarding course registration and academic activities. Since this process is handled manually, students often experience delays in getting responses, especially during busy periods. The lack of an automated system increases the workload of staff and affects the efficiency of communication.

PROPOSED AI SOLUTION:

The proposed solution is an AI-powered chatbot developed using Google Dialogflow. The chatbot understands user questions by identifying intents and entities and then provides appropriate responses. It offers instant assistance for course registration, elective courses, registration deadlines, credit limits, and other academic services, making the process faster and more convenient.





PROJECT COVERAGE :

The chatbot focuses on answering common questions related to course registration and academic activities. It provides information about registration procedures, deadlines, electives, credit limits, and course withdrawal. Although the chatbot currently works with predefined responses, it can be expanded in the future by integrating it with the university database.

CHATBOT ARCHITECTURE :

The chatbot works as an intelligent virtual assistant. It receives a user's query, identifies the user's intention using Natural Language Processing, and generates the most suitable response. Different training phrases help the chatbot recognize similar questions and improve its response accuracy.

DIALOGFLOW IMPLEMENTATION :

The chatbot is developed using Google Dialogflow ES. Various intents are created to handle different student queries, while entities help identify important information such as course names, semesters, and departments. Training phrases and responses are added for each intent to make conversations more effective.

CONVERSATION PROCESS :

The chatbot follows a simple interaction process:

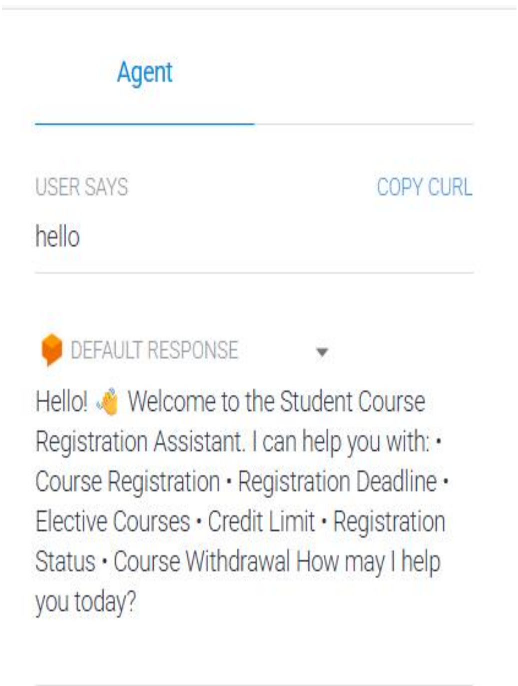
- ✓ User enters a question.
- ✓ Dialogflow identifies the intent.
- ✓ Entities are extracted if necessary.
- ✓ The chatbot generates the appropriate response.
- ✓ The conversation continues until the user's query is resolved.

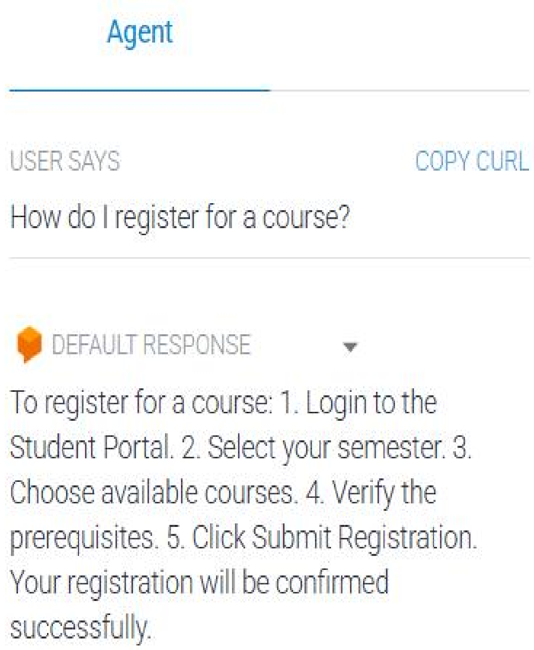
AI SEARCH TECHNIQUE:

Breadth-First Search (BFS) is used in this project to explain the concept of AI search techniques. BFS explores all possible options at the current level before moving to deeper levels. Although Dialogflow internally uses machine learning algorithms, BFS is included to demonstrate AI concepts required for the assessment.

PERFORMANCE EVALUATION :

The chatbot was tested using different user queries to verify whether it correctly identified the user's intent and generated the expected response. The chatbot responded accurately to most predefined questions and provided results within a short response time.





BENEFITS OF THE SYSTEM:

- ✓ Provides instant responses.
- ✓ Reduces manual work.
- ✓ Improves communication.
- ✓ Available 24×7.
- ✓ Easy to use.
- ✓ Can be upgraded in the future.

CHALLENGES AND LIMITATIONS:

Works only with predefined intents.

Cannot answer unknown questions.

Requires internet connectivity.

Needs regular updates.

Does not provide personalized information without database integration.

FUTURE IMPROVEMENTS:

The chatbot can be enhanced by connecting it with the university ERP system. Future improvements may include student login, attendance tracking, examination results, fee details, multilingual support, voice interaction, and mobile application integration.

CONCLUSION:

The **AI-Based Student Course Registration Assistant** successfully demonstrates the practical use of Artificial Intelligence in education. The chatbot provides quick, accurate, and reliable responses to common student queries while reducing the workload of administrative staff. The project also highlights the use of Dialogflow, NLP, intent recognition, entity extraction, and AI search techniques in developing an intelligent conversational system.

Evaluation Rubrics

Criteria	Weightage (%)	Excellent (Full Marks)	Good	Average	Poor
Problem Understanding & AI Concept Identification	20%	10 – Clearly analyzes the student support problem, defines AI application, agent objective, users, environment, inputs, outputs, and performance measures with proper justification.	7.5 – Identifies most components of the agent design with minor omissions.	5 – Provides basic problem analysis with limited explanation of agent components.	0–2.5 – Incomplete understanding of problem and agent design.
Dialogflow Agent Design & Implementation	30%	10 – Successfully creates a functional Dialogflow agent with appropriate intents, entities, training phrases, responses, contexts, and explains interaction flow with screenshots.	7.5 – Develops major Dialogflow components with minor configuration errors.	5 – Creates basic intents and responses with limited entity/context utilization.	0–2.5 – Incomplete or incorrect Dialogflow implementation.
Problem Formulation & Conversation Flow Design	20%	10 – Clearly represents chatbot interaction using initial state, goal state, actions, state transitions, goal conditions, and a well-designed conversation flow diagram.	7.5 – Provides a structured flow with minor missing details.	5 – Provides basic flow representation with limited explanation.	0–2.5 – Poor or missing problem formulation and decision flow.
Application of AI Techniques & Reasoning	20%	10 – Correctly analyzes and applies suitable search strategies (BFS/heuristic search) for intent selection or dialogue navigation with strong justification.	7.5 – Applies appropriate search strategy with reasonable explanation.	5 – Provides basic understanding of search strategy application.	0–2.5 – Unable to relate search concepts with conversational agent.
Testing, Evaluation & Documentation	10%	10 – Performs comprehensive testing using multiple user queries, evaluates accuracy, response quality, unknown query handling, and suggests meaningful improvements.	7.5 – Performs adequate testing and provides evaluation with minor gaps.	5 – Conducts limited testing with basic observations.	0–2.5 – Insufficient testing and evaluation.